Inbo Sim

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SUMMARY

PhD candidate in Electrical Engineering with a strong background in flexible electronics and system-level multimodal sensing with embedded chips (***proximity, imaging, tactile and force sensing***) for human perception and human-machine interaction.

EDUCATION

University of Virginia — **Charlottesville, VA**
- Doctor of Philosophy in Electrical Engineering — August 2023 – Present
  Anticipated graduation: 05/2027

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — **Daegu, South Korea**
- Bachelor of Engineering in Electrical Engineering — March 2018 – August 2023

RESEARCH EXPERIENCE

Intelligent Sensing Lab, University of Virginia — **Charlottesville, VA**
PhD candidate (Advisor: Professor Kyusang, Lee) — August 2023 – Present

- **Multimodal E-Skins with Haptic Interfaces for Augmented Human-Robot Interaction**
  - Fabricated and integrated flexible multimodal e-skin systems combining NIR-based proximity sensing and piezoresistive tactile arrays for human–robot interaction platforms
  - Developed heterogeneous integration of InGaAs photodiode arrays, GaAs VCSEL emitters, and MWCNT-based tactile sensors on flexible substrates for conformal robotic fingertips
  - Implemented sensor–actuator closed-loop system by synchronizing multimodal sensors with haptic interfaces, enabling real-time proximity mapping and tactile feedback for teleoperated robotics
  - Built shared autonomous platform to fuse supervisory input from user with precise robotic execution.

- **Reconfigurable Multi-Axis Force Recognition in 3D Space for Adaptive Human-Robot Interaction**
  - Developed a reconfigurable optical multi-axis force sensing system and fused it with robotic joint kinematics to enable 3D force reconstruction in operational space
  - Inferred human intent during interaction by correlating force patterns with motion trajectories, enabling adaptive human-robot collaboration
  - Enabled vectorized force sensing across five fingertips to support intention-aware control and adaptive motion planning for physical human-robot collaboration

- **Enhanced Human-machine collaboration through proximity sensing-based point cloud 3D image reconstruction**
  - Developed a flexible proximity sensing based scanning framework to reconstruct 3D point clouds in vision-limited environments.
  - Integrated NIR photodiode, VCSEL with edge computing chips through a flexible interposer to enable an edge computing.

- **Flexible pressure sensor integrated Oxide-based Thin Film Transistor with Piezoelectric material for Gait Analysis**
  - Designed and integrated piezoelectric pressure sensor with indium zinc oxide (IZO) thin film transistor and polyvinylidene fluoride trifluoroethylene (PVDF-TrFE).
  - Fabricated a flexible PCB (FPCB) for real-time pressure measurement and monitoring during gait analysis.

Energy Device and Nano-Engineering Lab, Seoul National University — **Seoul, South Korea**
Research Intern (Advisor: Professor Yun Seog, Lee) — January – February 2023

- Executed dry and wet transfer processes to deposit membrane-like thin-film materials onto target substrates, maintaining structural integrity and achieving consistent adhesion across substrate surfaces.
- Characterized transferred films to evaluate thickness uniformity, surface morphology and defect density,

ensuring material quality prior to device integration.

Thin Film Devices Laboratory, University of Virginia **Charlottesville, VA**
Research Intern (Advisor: Professor Kyusang, Lee) June – August 2022

- Participated in a flexible piezoresistive device project aimed at enabling early diagnosis of neurodegenerative disorders.
- Designed circuit on Cu-coated PI film and thin-film device using Autocad. Organized circuit using Arduino, Analog Discovery and logic devices for measurement of thin-film transistor and memristor

Bio-integrated Electronics Laboratory, DGIST **Daegu, South Korea**
Undergraduate Research Intern (Advisor: Professor Kyung-in, Jang) September 2021– May 2022

- Fabricated flexible devices using the process of semiconductor fabrication; LED patches applied to research of the brain depending on wave lengths, drug delivery devices using thermopneumatic micropump, electrocardiography (ECG) Circuit.
- Fabricated microneedle and microfluidics to allow drug delivery and integrated them into multimodality device.

Undergraduate Group Research Project Program, DGIST **Daegu, South Korea**
Undergraduate Research Intern (Advisor: Professor Kyung-in, Jang) January–December 2021

- Proposed platform to prevent infants from unpredictable death, called sudden infant death syndrome (SIDS).
- Designed and printed 3D modeling using 3D Shapr and Autocad to package every circuit board attached to the platform, increasing durability and fitness to their position on bed.
- Organized and analyzed the circuit to operate whole system consisting of vibration motor, LED and Bluetooth module to communicate with application our team created.

Multiscale Biomedical Robotics Laboratory, DGIST **Daegu, South Korea**
Research Intern (Advisor: Professor Suk Ho, Park) January–February 2021

- Executed basic experiment to demonstrate the feasibility of research about guide wire with Whole skin locomotion (WSL) to reduce the risk of damaging vessel by achieving smooth motion.
- Designed the prototype of guide wire and fabricated actual model using small size magnet and ferrofluid, leading to inducing Whole skin locomotion caused by magnetic forces.

PUBLICATION

1. **I. Sim***, B. Bae*, S. Hasan, Y, Baek, D. H., Seo, W. Cho, J. Liu, Y. M. Song, J. Heo, T. Iqbal, S. Heo, K. Lee, "Multimodal Sensory E-skin and Synchronized Haptic Interfaces for Augmented Human-Robot Interaction" ***Nature Electronics*** (Accepted), *These authors contributed equally to this work.
2. B. Bae*, S. Kim*, **I. Sim***, S.B. Choi*, N. Bezzo, S. Hong, I. Lee, T. Iqbal, S.Chung, F. Fioretto, J. Kung, K. Lee, "Embodied edge computing for real-time perception in physical AI" ***Nature Reviews Computing*** (Accepted) *These authors contributed equally to this work.
3. Y. Baek, B. Bae, D. H. Seo, J. Cho, **I. Sim**, D. Kim, S. Kim, Y. Gong, S. Kim, D. Kim, Y. H. Jeon, S. Chung, S.-H. Song, Y. M. Song, Y. S. Lee "Personalizable gait sensing through meta-learned on-chip enrollment" (*under review*)
4. S. Jeong*, H. W. Ko*, J. Kang*, J. Ko*, **I. Sim***, Z. Xu, H. Jeong, Y. Choi, S. Jo, J. W. Shim, P. H. Seo, M. S. Chae, Y. Kim, H. Yeon, Y. Suh, H. Lee, J. Kim, K. Lee†, S. Bae†, M. Park†, S. Mun†, "Event-Driven Neuromorphic Gaze Decoding via e-Skin Electrooculography", ***ACS Nano*** (2026)
5. **I. Sim**, K. Choi, Y. Baek, J. H. Choi, J. Jo, J. Yeom, B. Kim, Y. Cho, H. Lee, H. Bang, J.-H. Han, D. H. Park†, J. Kim†, K. Lee†, "Microdisplay technologies in augmented reality and virtual reality headsets", ***Nature Review Electrical Engineering*** (2025)

6. Y. Baek*, B. Bae*, J. Yang, W. Cho, **I. Sim**, G. Yoo, S. Chung, J. Heo†, K. Lee†, "Trust Region Bayesian Optimized Network of Artificial Olfactory Receptors for Spatiotemporal Monitoring of Toxic Gas", ***Science Advances*** (2024), *These authors contributed equally to this work.
7. B. Bae*, M. Park*, D. Lee*, Y. Baek, Y. Mu, S. Kim, T. Kim, J. Yang, **I. Sim**, C. Shen, K. Lee†, "Stereoscopic Artificial Compound Eyes for Spatiotemporal Perception in Three-dimensional Space", ***Science Robotics*** (2024) *These authors contributed equally to this work. *Selected as the paper of the year in 2024.*
8. B. Bae*, M. Park*, D. Lee*, **I. Sim**, K. Lee†, **"**Hetero-Integrated InGaAs Photodiode and Oxide Memristor-Based Artificial Optical Nerve for In-Sensor NIR Image Processing**",** ***Advanced Optical Materials*** (2023) *These authors contributed equally to this work.

TECHNICAL SKILLS

- Hands-on experiences in microfabrication and electronics devices such as NIR photodiode, VCSEL, piezoresistive and piezoelectric material and thin-film transistor (IGZO / IZO).
- Extensive hands-on experience with photolithography techniques, including laser direct writing and mask alignment.
- Proficient in thin-film deposition process such as atomic layer deposition (ALD), sputtering, electron beam evaporation, plasma-enhanced chemical vapor deposition (PEVCD) and electrochemical plating.
- Knowledgeable in wet and dry etching methods such as reactive ion etching (RIE).
- Skilled in executing bonding processes with flip-chip bonder and wafer bonder equipment for heterogeneous integration.
- Experienced in designing printed circuit board (PCB) and customized flexible PCBs for electrical characterization of fabricated devices.
- Proficient in surface and optical characterization tools, including profilometers, ellipsometers.

TEACHING EXPERIENCE

- **Teaching Assistant,** *From AI to Nano (ENRG 2500)*, University of Virginia — Spring 2026
- **Teaching Assistant,** *From AI to Nano (ENRG 2500)*, University of Virginia — Spring 2025

SERVICE

Military Service, United States Army Garrison Humphreys — **Pyeongtaek, South Korea**
Sergeant in Headquarters and Headquarters Company — March 2019 – August 2020

- Reviewed Operation Orders from United States Force Korea as a current Operational Specialist and arranged them to support the largest overseas military base, United States Army Garrison, Humphreys.
- Processed daily requests such as leave/pass, awards and personal action in timely manner for 100+US Soldiers assigned to the Garrison as administrative specialist.